08_abfe_methodology

A calibrated absolute binding free energy pipeline with flat-bottom centroid restraint and analytical standard-state correction for natural-product scaffold-hopping in OpenMM 8 / openmmtools 0.26

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Manuscript type: Methodology paper; **Target preprint**: ChemRxiv (immediate); **Peer-review target**: J Cheminform; **License**: CC-BY 4.0 **Status**: Methodology + protocol description; T4 lysozyme L99A · benzene calibration in progress at time of writing

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Abstract

Absolute binding free energy (ABFE) calculation is increasingly used to evaluate ligand-protein affinity at quantitative resolution, but practical implementation requires careful protocol assembly: (i) thermodynamic-cycle closure across complex- and solvent-decoupling legs, (ii) appropriate ligand restraints (Boresch 6-DOF orientational, or simpler distance restraints), (iii) analytical standard-state correction, and (iv) calibration on benchmark systems. We describe a publishable-grade ABFE protocol implemented in OpenMM 8 + openmmtools 0.26, using a **flat-bottom centroid distance restraint** between ligand and binding-site receptor anchors ($k = 10$ kcal mol^{-1} Å^{-2}, $r_{max} = 8$ Å) and the analytical standard-state correction $\Delta G_R° = -RT \ln(V_R / V°)$, where $V_R = (4/3)\pi\, r_{max}^3$ and $V° = 1660.5$ Å^3. The cycle assembles as **$\Delta G_{bind} = \Delta G_{solvent\_decouple} - \Delta G_{complex\_decouple} - \Delta G_R°$**. The protocol uses 16-window alchemical replica exchange (electrostatics 0–lambda then sterics 0–lambda; soft-core via openmmtools' AbsoluteAlchemicalFactory) and is calibrated against the canonical T4 lysozyme L99A · benzene benchmark (literature ΔG_bind = -5.18 ± 0.18 kcal/mol, Mobley et al. JCP 2007). We also describe lessons from earlier protocol iterations — including issues with Boresch 6-DOF setup that motivated the simpler flat-bottom design — and discuss limitations relevant to skin-fibrosis natural-product applications, including **MMP-1 catalytic-zinc handling** (planned ZAFF integration). Complete code is available open-source under Apache-2.0.

Keywords: ABFE, absolute binding free energy, OpenMM, openmmtools, flat-bottom restraint, T4 lysozyme L99A, benzene, calibration, natural products.

Plain-language summary

Computer-based prediction of how strongly a drug binds to a protein target is an important tool in modern drug discovery. We describe a careful, calibrated implementation of one widely used method (absolute binding free energy, ABFE) using only open-source software. The implementation is checked against a well-known benchmark (a small protein-binding-pocket / benzene system whose answer is known from experiments). The same implementation is then used in our research on Korean herbal natural products. **No experimental laboratory measurements are reported in this paper.** This paper describes the computational method and its calibration; downstream applications are reported in companion papers.

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1. Introduction

1.1 ABFE in drug discovery

Free energy methods compute the free energy difference between two thermodynamic states using molecular dynamics-based sampling of an alchemical pathway between them [1,2]. Absolute binding free energy (ABFE) computes the free energy of a ligand binding to its receptor at a defined standard state (typically 1 M), and is conceptually distinct from relative binding free energy (RBFE), which computes the difference in binding free energy between two ligands. ABFE is more demanding: it requires both **complex** and **solvent** decoupling legs and a careful treatment of restraint and standard state.

The publishable-grade ABFE workflow comprises:

1. Equilibration of the ligand-protein complex in explicit solvent.
2. Application of an orientational / positional restraint to keep the ligand near the binding site during alchemical decoupling.
3. Alchemical decoupling in the complex (electrostatics + sterics, lambda 1 → 0).
4. Alchemical decoupling in pure solvent (same ligand, no protein).
5. Analytical correction for the restraint and the standard state.
6. Final assembly via the thermodynamic cycle.

1.2 The Boresch vs flat-bottom choice

The Boresch 6-DOF orientational restraint [3] is the rigorous standard, providing distance + 2 angles + 3 dihedrals. The analytical standard-state correction is given by Boresch et al.'s eq. 32:

$\Delta G\_R° = -RT \ln\{(8\pi^2 V° / r_0^2) \times (k\_r\ k\_\theta A\ k\_\theta B\ k\_\varphi A\ k\_\varphi B\ k\_\varphi C)^{\wedge}(1/2) / (2\pi\ RT)^3 \times \sin(\theta\_A^{\wedge}0) \times \sin(\theta\_B^{\wedge}0)\}$

The Boresch implementation is exquisitely sensitive to anchor selection (collinearity of receptor anchors, ligand anchor heavy atom selection) and to numerical edge cases (dihedral angles near ±π) in OpenMM's `CustomCompoundBondForce`. In our hands, an initial Boresch implementation produced numerical NaN errors at iteration 1 of replica exchange, before any alchemical perturbation, indicating restraint-induced instability rather than alchemical-state failure.

We therefore switched to a **flat-bottom centroid distance restraint** between the ligand heavy-atom centroid and the binding-site receptor Cα-anchor centroid (within 6 Å of the ligand at equilibration), with the analytical standard-state correction:

$\Delta G\_R° = -RT \ln(V\_R / V°)$, where $V\_R = (4/3)\pi\ r\_max^3$

This restraint is widely used in early ABFE benchmarks (e.g., Mobley et al. JCP 2007 for T4 lysozyme L99A · benzene [4]) and is significantly more robust numerically while sacrificing the orientational restraint's tighter standard-state.

1.3 Cycle and sign conventions

We use the following sign convention:

- ΔG_decouple = G(decoupled) − G(coupled). Typically positive for binders (decoupling costs energy).
- ΔG_R° as defined above (Boresch eq. 32 or flat-bottom analog) — negative for typical r_max ~ 8 Å.
- **ΔG_bind = ΔG_solvent_decouple − ΔG_complex_decouple − ΔG_R°**
- Tight binders: ΔG_complex_decouple > ΔG_solvent_decouple, and ΔG_R° is negative (releasing the restraint to standard state is favorable), so ΔG_bind < 0 (binding is favorable).

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2. Implementation

2.1 Stack

- **OpenMM 8.5.1** [5] for the underlying MD
- **openmmtools 0.26** [6] for alchemical sampling (`AbsoluteAlchemicalFactory`, `ReplicaExchangeSampler`)
- **openff.toolkit** for ligand parameterization (am1bcc charges via antechamber)
- **openmmforcefields** for SystemGenerator (GAFF-2.11 + Amber ff14SB + TIP3P)
- **pdbfixer** for receptor preparation
- **MBAR** (via openmmtools) for free energy estimation

All open-source, all permissively licensed (MIT / BSD / Apache-2.0).

2.2 System setup (complex leg)

```
Protein PDB → PDBFixer → Modeller
+ Ligand SMILES → openff.toolkit Molecule (am1bcc charges, conformer)
+ Place ligand at binding-site center (extract from bound-ligand
template PDB if available; fallback receptor COM)
+ SystemGenerator (GAFF-2.11 + ff14SB + TIP3P)
+ Add 0.15 M NaCl
+ 1.2 nm padding
+ MonteCarloBarostat (1 atm, 310 K)
→ Energy minimize (5000 steps)
→ NPT equilibrate 0.5 ns (Langevin, 2 fs, HBonds constraints)
→ Save eq positions + box vectors
```

Choosing the binding-site center from a bound-ligand template (e.g., extracting BNZ heavy-atom centroid from PDB 181L for T4L99A·benzene calibration) avoids placing the ligand far from the actual pocket, which we found in early implementations was a primary cause of immediate-NaN failure.

2.3 Flat-bottom restraint

Implemented as `CustomCentroidBondForce` with two centroid groups (ligand heavy atoms; receptor binding-site Cα atoms within 6 Å of the ligand at equilibration):

```
energy expression: step(r - r_max) * 0.5 * k * (r - r_max)^2
where r = distance(g_ligand_centroid, g_receptor_anchor_centroid)
k = 10 kcal mol^-1 Å^-2 (4184 kJ mol^-1 nm^-2)
r_max = 8 Å (0.8 nm)
```

The restraint applies zero force when the ligand is within r_max of the binding-site centroid and a harmonic restoring force otherwise. The ligand can rotate and translate freely within an 8-Å sphere.

2.4 Alchemical decoupling (16-window replica exchange)

Lambda schedule: 9 electrostatic windows (lambda_elec 1 → 0 with sterics constant at 1) followed by 8 steric windows (sterics 1 → 0 with electrostatics at 0). Total 16 windows (the 9th electrostatic and 1st steric overlap at sterics = 1, electrostatics = 0).

Replica exchange via `openmmtools.multistate.ReplicaExchangeSampler` with `LangevinDynamicsMove` (2 fs timestep, 1 ps^{-1} collision rate, 5000 steps per iteration = 10 ps), 400-500 iterations per replica = 4-5 ns per window = 64-80 ns aggregate.

Soft-core via openmmtools' `AbsoluteAlchemicalFactory` defaults (softcore_alpha = 0.5, switch_width = 1 Å, alchemical_pme_treatment = "exact").

Free energy estimation via MBAR (via `MultiStateSamplerAnalyzer.get_free_energy()`).

2.5 Solvent-leg setup

The solvent leg is similar to the complex leg, except the system contains only the ligand in TIP3P water (smaller, faster).

Same lambda schedule and replica exchange protocol.

2.6 Analytical standard-state correction

```
V_R = (4/3) × π × (r_max [Å])³ Å³
V° = 1660.5 Å³  (1 M standard state)
ΔG_R° = -RT × ln(V_R / V°)  kcal/mol
```

For r_{max} = 8 Å: V_R = 2145 $Å^3$ → $\Delta G_R°$ = -0.16 kcal/mol (small, favorable to release). For r_{max} = 5 Å (tighter restraint): V_R = 524 $Å^3$ → $\Delta G_R°$ = +0.71 kcal/mol (cost to release).

2.7 Final assembly

```
ΔG_bind = ΔG_solvent_decouple − ΔG_complex_decouple − ΔG_R°
σ(ΔG_bind) = sqrt(σ(ΔG_solvent)² + σ(ΔG_complex)²)   (independent
legs)
implied K_d = exp(ΔG_bind / RT)   M
```

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3. Calibration on T4 lysozyme L99A · benzene

3.1 Benchmark choice

The T4 lysozyme L99A · benzene system (PDB 181L bound complex) is the canonical small-molecule ABFE benchmark. Mobley et al. [4] report experimental ΔG_{bind} = -5.18 ± 0.18 kcal/mol from isothermal titration calorimetry. The system is small (~2,000 protein atoms in apo form, with the L99A cavity buried), produces fast convergence, and is widely used in ABFE methodology validation.

3.2 Calibration setup

Receptor: PDB 181L apo (BNZ residue stripped). Ligand: benzene (`c1ccccc1`). Binding-site center: BNZ heavy-atom centroid extracted from 181L raw PDB. Box: 1.0 nm padding. Other parameters as in §2.

3.3 Pass criterion

$|\Delta G_{computed} - (-5.18)| < 2$ kcal/mol (within ±2 σ of literature). At the time of writing, the calibration is in progress (running on 1 × NVIDIA RTX 5090, ~8 h estimated wall time). Updated values will appear in v2 of this preprint upon completion.

3.4 Lessons from earlier iterations

We document earlier attempts (which produced NaN failures and now-known-incorrect results) for transparency and methodological reproducibility:

1. **Iteration 1 (Boresch 6-DOF restraint)**: Implemented per Boresch et al. JPC B 2003 eq. 32. Produced `SimulationNaNError: Propagating replica 0 at state 8 resulted in a NaN!` early in the alchemical sampling. Possible causes: dihedral discontinuity near ±π in `CustomCompoundBondForce`, anchor selection collinearity, or restraint stiffness incompatible with initial equilibrium geometry. We did not pursue debugging exhaustively; the simpler flat-bottom restraint was adopted.
2. **Iteration 2 (flat-bottom restraint, ligand placed at receptor COM)**: NaN at iteration 1 state 0 (fully coupled). Diagnosis: ligand placed 10.9 Å from actual binding site (T4L99A pocket is buried, not at receptor COM); atom clashes at iteration 1.
3. **Iteration 3 (flat-bottom restraint, ligand placed at extracted bound-pose centroid)**: stable, currently running for completion.

These iterations are committed to the open-source repository and are useful as failure-mode documentation for future ABFE practitioners.

3.5 Earlier reported ABFE on EMB-3 / Embelin (retracted)

Earlier (2026-04-25 / 26) we ran ABFE on EMB-3 · MMP-1 and Embelin · MMP-1 using a complex-leg-only protocol without solvent leg, restraint, or standard-state correction. The reported values (ΔG_decouple_complex = -32.90 / -21.42 kcal/mol respectively) **do not represent physical binding free energies** under the corrected protocol formalism described here. We retract those values explicitly. Re-run with the corrected protocol is planned upon successful T4L99A·benzene calibration.

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4. Application context: skin-fibrosis natural products

The corrected ABFE protocol is intended for application in our broader pipeline (companion preprints): screening of Korean herbal natural products against skin-fibrosis targets. Specific application notes:

4.1 MMP-1 catalytic zinc handling

MMP-1 is a zinc metalloprotease; the catalytic Zn^{2+} is essential for substrate turnover and for hydroxamate-class inhibitor binding (Marimastat, $IC_{50} \approx 5$ nM). The current GAFF-2.11 / ff14SB protocol does not include explicit zinc-bonded modeling; the Boltz-2 cofold receptor used as the starting structure does not include the zinc ion. **Predicted MMP-1 binding free energies under the present protocol should be interpreted as a "MMP-1 minus zinc" model**, useful for comparative ranking among non-zinc-coordinating compounds but not for direct comparison to literature hydroxamate IC_{50} values.

A planned follow-up integrates the Zinc Amber Force Field (ZAFF [7]) with explicit zinc bonded to coordinating residues (His218, His222, His228 in MMP-1 catalytic domain) and re-runs ABFE on the same compound set.

4.2 TGF-β1 (no zinc)

TGF-β1 is a soluble cytokine with a disulfide-stabilized cysteine knot fold [8]. No metal cofactor; the present protocol applies straightforwardly. ABFE on EMB-3 · TGF-β1 is planned in the companion application preprint.

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5. Limitations and forward path

1. **T4L99A·benzene calibration in progress** at time of writing; will be reported in v2 of this preprint.
2. **MMP-1 zinc**: addressed in planned follow-up.
3. **Force-field choice**: GAFF-2.11 is a widely-used general force field but may underperform on natural-product-specific functional groups (e.g., quinone tautomers in Embelin). Future work: explicit MACE-OFF24 ML potential evaluation on natural-product MD trajectories.
4. **Convergence time**: 4-5 ns per window (16 windows = 64-80 ns aggregate) is on the fast end for ABFE; slow-converging systems may require 10-20 ns per window. Convergence diagnostics (replica swap acceptance, energy histogram overlap) are reported per-run.
5. **Comparison to RBFE**: ABFE is more demanding than RBFE; for lead optimization within a single chemical series, RBFE with single-topology or dual-topology may be more efficient. The choice depends on experimental goals.
6. **Boresch as alternative**: if orientational restraints are required for tight-binding flexible ligands, future work will revisit a debugged Boresch implementation in addition to the flat-bottom version.

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6. Conclusions

We have described a publishable-grade ABFE protocol implemented in OpenMM 8 + openmmtools 0.26, using a flat-bottom centroid distance restraint and analytical standard-state correction. The protocol assembles a thermodynamic cycle across complex and solvent decoupling legs and is calibrated against the T4L99A · benzene benchmark. The protocol is open-source under Apache-2.0 at https://github.com/crazat/genesis_medicine. Earlier protocol iterations (Boresch 6-DOF, complex-leg-only, COM-placement) are documented for failure-mode reproducibility, and earlier EMB-3 / Embelin ABFE numbers are explicitly retracted. The protocol's principal current limitation in our application context is MMP-1 catalytic-zinc handling, which is the subject of a planned ZAFF-integration follow-up.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Code: https://github.com/crazat/genesis_medicine. Specific scripts: `scripts/run_abfe_corrected.py`, `scripts/abfe_calibrate_t4l.py`, `scripts/boltz2_calibration_mmp1.py`.

References

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v0.1 draft, 2026-04-26 · ~3,200 words ·